

C in One Shot

Part – 1

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Flow – 1

- 1) Basic Printing, \n wali cheez, printing numbers also with + and -.
- 2) Variables, printing variables, int, float, and +,-,*,/ of integers.
- 3) Variables naming rules.
- 4) Comments
- 5) Taking Input
- 6) Modulus Operator
- 7) Float to int, int to float
- 8) Hierarchy
- 9) Char and ASCII

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Basic program in C

```
#include<stdio.h>

int main(){

    printf("hello world");

    return 0;

}
```

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How to move in next line?

Example :

```
printf("Hello PW");  
printf("Hello CW");
```

Output will be :

Hello PWHello CW

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Use of escape sequence '`\n`'

Example :

```
printf("Hello PW");  
printf("\n");  
printf("Hello CW");
```

Output will be :

```
Hello PW  
Hello CW
```

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Use of escape sequence '\n'

Predict the output :

```
main(){
    printf("nn\n\nnn\n");
    printf("nn/n/nnn/n");
}
```

```
# include <stdio.h>
int main(){
    return 0;
}
```

Output

- nn
-
- nn
- nn/n/nnn/n

Variables and their Declaration

Let us focus on int data type as of now.

1) Variables as containers :

```
# include <stdio.h>
```

```
int main()
```

```
{
```

```
printf("Hello");
```

```
int n;
```

```
n = 3;
```

```
n = 5;
```

```
printf("%d", n)
```



Variables

↓

`n, y`

Literals

`'\n'`

`'%d'`

```
printf("n");
```

→ text

```
printf("%d", n);
```

Printing Variables in C & Updation of Variables

```
int x = 5;
```

→ $int\ n;$
 $n = 5;$

-7
 n

```
printf("%d", x);
```

```
x = 7;
```

```
printf("%d", x);
```

```
x = x + 6;
```

```
printf("%d", x);
```

```
x = x - 20;
```

→ $n = 13 - 20$

```
printf("%d", x);
```

$n = -7$

Output

5 7 13 -7

Arithmetic operations on int data type

```
int x = 5;
```

```
int y = 2;
```

```
printf("%d", x+y);
```

```
printf("%d", x-y);
```

```
printf("%d", x*y);
```

```
printf("%d", x/y);
```

'+' '-'

%d → int

$$5/2 = 2$$

```
int x = 4;
```

```
int y = 2;
```

```
int z = x + y;
```

$$\boxed{4}_x$$

$$\boxed{2}_y$$

$$\boxed{6}_z$$

$$\begin{array}{r} 2 \\ 2 \overline{) 5} \\ \underline{4} \\ 1 \end{array}$$

$5/2 \rightarrow 2.5$
 $\text{int } x = 5/2 ;$

$5/2 \rightarrow 2$

Increment – Decrement operators

```
int x = 5;  
x++;  
printf("%d",x);  
x--;  
printf("%d",x);  
++x;  
printf("%d",x);  
--x;  
printf("%d",x);
```

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Float data type

→ Real Numbers

```
float x = 3.1;
```

$\%d \rightarrow \text{int}$

$\%f \rightarrow \text{float}$

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Arithmetic operations on float data type

```
float x = 5;
```

```
float y = 2;
```

```
printf("%f", x+y);  $\rightarrow 7$ 
```

```
printf("%f", x-y);  $\rightarrow 3$ 
```

```
printf("%f", x*y);  $\rightarrow 10$ 
```

```
printf("%f", x/y);  $\rightarrow 2.5$ 
```

```
float n;  
n = 5;
```

5.00
n

$5/2 \rightarrow \text{integer} \rightarrow 2$

$\text{float } z = 5/2;$

$\rightarrow \text{float } z = 2;$

$5.0/2 \rightarrow 2.5$

$5/2.0 \rightarrow 2.5$

$$V = \frac{4}{3} \pi r^3$$

$$V = 4 * 3.14 * r * r * r / 3$$

int r=5

float v = 4 * 3.14 * 5 * 5 * 5 / 3

H.W./C.W. WAP to display area of circle with given radius.

Example : Calculating percentage of 5 subjects

```
float x1 = 90; // x1 can be physics
float x2 = 91; // x2 can be chemistry
float x3 = 92; // x3 can be maths
float x4 = 93; // x4 can be english
float x5 = 94; // ohh wait comments ke baare me to bataya hi nahi xD
float percent = (x1 + x2 + x3 + x4 + x5)/5;
printf("%f",percent);
// change the marks and run each time
```

H.W. Print percentage of 4 subjects whose marks are out of 40.

Example : Calculating Area of a Circle H.W.

```
float radius = 5;
```

```
float r = 5;
```

```
float pi = 3.1415;
```

```
float area = pi*radius*radius;
```

```
int x, int y int z
```

```
printf("%f",area);
```

```
int a
```

Variable Naming rules

- 1) Variables can start from an alphabet or underscore _ .
- 2) Special characters except _ are not allowed.
- 3) Some particular keywords are not allowed.
- 4) Commas or blanks are not allowed.

auto
~~Auto~~ double int break extern enum unsigned while
 case sizeof for const static long continue float
 else signed do short switch char volatile default
 goto struct if union return void register typedef

float macbook;

int Mac;

Variable Naming rules – Examples

Q. Which of the following are invalid variable names and why?

BASICSALARY

_basic

basic-hra

#MEAN

group.

422

population in 2006

over time

mindovermatter

FLOAT

hELLO

queue.

team'svictory

Plot#3

2015_DDay

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Example : Calculating Simple Interest

```
float p,r,t,si;
p = 100;
r = 10;
t = 2;
si = (p*r*t)/100;
printf("%f",si);
```

$$SI = \frac{PRT}{100}$$

$$\frac{100 \times 10 \times 2}{100}$$

$$\frac{2000}{10} = 200$$

Taking input // Let us take a simple example from User.

```
int x;

printf("Enter a number\n");

scanf("%d",&x); // user will give 'x' a value.

int y = x*x;

printf("square of number that you gave is %d",y);
```

```
#include<stdio.h>
int main(){
    float radius;
    scanf("%f",&radius);
    float pi = 3.1415;
    float area = pi * radius * radius;
    printf("The area of circle is : %f",area);
    return 0;
}
```

Run

10

The area of circle is : 314.15

Dry Run:

10
radius

3.1415
pi

314.15
area

Output

10

The area of circle is : 314.15

```
#include<stdio.h>
int main(){
    float principal,rate,time,si; // 4 dabbe c
    printf("Enter Principal : ");
    scanf("%f",&principal);
    printf("Enter Rate : ");
    scanf("%f",&rate);
    printf("Enter Time : ");
    scanf("%f",&time);
    si = (principal*rate*time)/100;
    printf("Your simple interest is : %f",si);
    return 0;
}
```

Output :

Enter Principal : 100

Enter Rate : 20

Enter Time : 3

Your simple interest is : 60

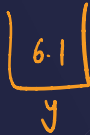
scanf ("%d", & n)



```
int n = 5;
```



```
float y = 6.1;
```



$\&n \rightarrow$ address of n

```
int n;
```

```
scanf("%d", &n);
```



Taking input // SUM of 2 given numbers

Homework:

```
int x;
printf("Enter first number\n");
scanf("%d",&x); // user will give 'x' a value.
int y;
printf("Enter second number\n");
scanf("%d",&y); // user will give 'y' a value.
int sum = x+y;
printf("sum of the numbers that you gave is %d",sum);
```

9
Sum

Output

Enter 1st No.

En — —

Taking Input

Predict the output :

```
main(){
    int p,q;
    printf("Enter values of p and q");
    scanf("%d %d ", &p, &q);
    printf ("p = %d q = %d", p, q);
}
```

30	40
p	q

Output

Enter Values of p & q

30 40

p = 30 q = 40

'Dry Run'

Example : Take two integers input, **a** and **b** : **a > b**, and find the remainder when **a** is divided by **b**. & print the remainder.

```
main() {
```

```
    int a, b;
```

```
    printf("Enter 1st No. ");
```

```
    scanf("%d", &a);
```

```
    printf("Enter 2nd No. ");
```

```
    scanf("%d", &b);
```

```
    int q = a / b;
```

```
    int r = a - (b * q);
```

```
    return 0;
```

```
}
```

$$\begin{array}{r} 5 \\ 7 \overline{) 39} \\ \underline{35} \\ 4 \end{array}$$

$$7 \times 5 + 4 = 39$$

```
int a = 39;
```

```
int b = 7;
```

```
int q = a / b; // q = 5
```

Divisor $\times$ Quotient + Rem = Dividend

$\Rightarrow$ Remainder = Dividend - Divisor $\times$ Quotient

$\Rightarrow r = a - (b \times q)$

Operators:

$+$, $-$, $/$, $*$, $\%$

Modulo Operator $\rightarrow$ $\%$

$a = 2;$

$b = 6;$

$$2 \% 2 = 2$$

$$a \% b = a$$

$$[n > 2]$$

$$[a < b]$$

$$r = a \% b$$

$$\frac{2}{6} = 0$$

$$38 \% 8 = 6$$

$$\begin{array}{r} 4 \\ 8 \overline{) 38} \\ \underline{32} \\ 6 \end{array}$$

float to int and int to float :

Ques : Take integer as input and print half of the number.

↗ positive

Ques : Take float input and print the **fractional** part of the real number.

$$\{x\} = x - [x]$$

1, 2, 3, 4, 5, 6

'3.14' → 0.14

G.I.F.

$[x]$

$$x = 7.1$$

$$[x] = 7$$

'5.7' = 5 + 0.7

0.7



```
float x; →
```

```
scanf("%f", &x);
```

```
int y;
```

```
y = x;
```

```
float z = x - y;
```

float
 int

5.7

x

5

y

0.7

z

Output

0.7

Homework: Code it Yourself

Hierarchy of operators

BODMAS →

B, O, D/M, A/S

```
int i = 2 * 3 / 4 + 4 / 4 + 8 - 2 + 5 / 8 ;
printf("%d", i);
```

PROG

$$(2 * 3) / 4 = 6 / 4 = 1$$

$$2 * (3 / 4) = 2 * 0 = 0$$

$$i = 1 + 1 + 8 - 2 + 0 = 8$$

8

MATHS

$$(2 * 3) / 4 = 6 / 4 = 1.5$$

$$2 * (3 / 4) = 2 * 0.75 = 1.5$$

Try This!

Predict the output :

```
int main(){
```

```
    ✓ int i = 2, j = 3, k, l;
```

```
    float a, b;
```

```
    k = i / j * j;  $(2/3) * 3 = 0 * 3 = 0$ 
```

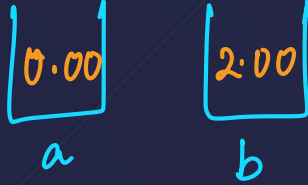
```
    l = (j / i) * i;  $= (3/2) * 2 = 1 * 2 = 2$ 
```

```
    a = (i / j) * j;  $(2/3) * 3 = 0 * 3 = 0$ 
```

```
    b = (j / i) * i;  $= 2$ 
```

```
    printf("%d %d %f %f", k, l, a, b);
```

```
}
```



Output

```
0 2 0.000000
2.000000
```

** Try This!

Predict the output :

```
int main(){
    float a = 5, b = 2;
    int c;
    c = a % b;
    printf("%d", c);
}
```

$a = 5.000000$
 $b = 2.000000$



1X

int → +, -, *, /, %
 float → +, -, *, /

Output

0, Error

char data type

char **ch** = 'a';
 ↗ variable
 ↘ %c

'a'

↗ n

char n = 'a';

'a', 'b', 'c', ...

'A', 'B', 'C', ... 'Z'

int, float
 ↗ integer
 ↘ real number/decimal
 ↘ %f
 ↘ %d
 ↘ format specifier

char ch = '#';

↗
 '#'
 ch

ASCII values

→ Important topic

→ good questions

```
char ch = 'a';
```

A → 65

B → 66

C → 67

D → 68

E → 69

F → 70

G → 71

z = 90

a = 97

b = 98

c = 99

d = 100

z = 122

→ 35

! → 33

+ →

=

_

-

@ → 64

* → 42

/

\$ → 36

If - Else

MCQ Time !

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MCQ 1

Character $\rightarrow$ single digit

'a', 'b' '#'

char ch = 'a#';

Which of the following is NOT a character constant

- (1) 'Thank You' WRONG
- (2) 'Enter values of P, N, R'
- (3) '23.56E-03'
- (4) ✓ All the above

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MCQ 2

In $(b = 6.6 / a + 2 * n;)$ which operation will be performed first?

☒ (1) $6.6 / a$

☐ (2) $a + 2$

☐ (3) $2 * n$

☐ (4) Depends upon compiler

MCQ 3

Which of the following statements is false

- (1) Each new C instruction has to be written on a separate line
- (2) Usually all C statements are entered in small case letters
- (3) Blank spaces may be inserted between two words in a C statement
- (4) Blank spaces cannot be inserted within a variable name

```
int n = 4;  
n = n + 7;
```

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MCQ 4

If a is an integer variable, $a = 5 / 2$; will return a value

(1) 2.5

(2) 3

☒ (3) 2

(4) 0

`int a = 5/2;`

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MCQ 5

BODMAS

The expression, $a = 7 / 22 * (3.14 + 2) * 3 / 5$; evaluates to

(1) 8.28

(2) 6.28

(3) 3.14

☒ (4) 0

$$\downarrow$$

$$7/22 * 5.14 * 3/5$$

$$0 * 5.14 * 3/5$$

$$0 * 3/5$$

$$0/5$$

$$\downarrow$$

$$0$$

MCQ 6

The expression, ^{int} $a = 30 * 1000 + 2768$; evaluates to

☒ (1) 32768

$$30000 + 2768$$

(2) -32768

$$a = 32768$$

(3) 113040

(4) 0

$$\text{Short } a = 32768;$$

$$\curvearrowright - 32768$$

Data Types

$$2^{16} = 2^{10} \cdot 2^6$$

$$= 1024 \rightarrow 64$$

int v/s Short $\rightarrow$ Chhota dabba

int x = 3;

float y = 3.14;

char ch = 'A';

short z =

long x = 32768;

long long x =

-32768 to +32767 2^{15}

int $\rightarrow$

Short $\rightarrow$ 2 bytes $\rightarrow$ 16 bits

8 Bits $\rightarrow$ 1 byte

n bits $\rightarrow 2^n$ numbers

16 bits $\rightarrow 2^{16}$ numbers store

$$2^{16} = 65536$$

b Kb Mb Gb Tb

1000 1000 1000 1000

int data type $\rightarrow$ 4 bytes $\rightarrow$ 32 bits

$\rightarrow 2^{32}$ numbers
 $\downarrow$

long $\rightarrow$ 8 bytes $\rightarrow$ 64 bits
 2^{64} numbers

short -2^{15} to $2^{15} - 1$

int -2^{31} to $2^{31} - 1$

long long -2^{63} to $2^{63} - 1$

Char $\rightarrow$ 1 byte $\rightarrow$ 8 bits $\rightarrow$ 2^8 numbers $\rightarrow$ 256 numbers
 $\downarrow$

MCQ 7

BODMAS $\rightarrow$ $\%$ $/, \times$ $+, -$

The expression $x = 4 + 2 \% - 8$ evaluates to

(1) -6

$$\Rightarrow x = 4 + 2 = 6$$

☒ (2) 6

(3) 4

(4) None of the above

$$2 \% 8 = 2$$



$$8 \overline{) 2}$$

$$2 \% - 8 = 2 \% 8 = 2$$

$$5 \% 2$$

$$\begin{array}{r} 2 \overline{) 5} \\ \underline{4} \\ 1 \end{array}$$

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MCQ 8

What will be the value of d if d is a float after the operation $d = 2 / 7.0$?

- (1) 0
- (2) 0.2857
- (3) Cannot be determined
- ☒ (4) None of the above

$$\begin{array}{r} \text{float } d = 2/7.0; \\ 0.285714 \\ 7 \overline{) 2.0} \\ \underline{14} \\ 60 \\ \underline{56} \\ 40 \\ \underline{35} \\ 50 \\ \underline{49} \end{array}$$